

sequenceDiagram participant User participant Frontend as PWA Frontend participant Backend as FastAPI Backend participant ZKSystem as Zero-Knowledge System participant Blockchain as Mina Blockchain participant P2P as Torrent Network

%% User Registration Flow

User->>Frontend: Register/Login
Frontend->>Backend: Request ZK Identity Setup
Backend->>ZKSystem: Generate ZK Parameters
ZKSystem->>Frontend: Return ZK Parameters
Frontend->>Frontend: Generate ZK Proof (Client-side)
Frontend->>Backend: Submit ZK Proof
Backend->>ZKSystem: Verify ZK Proof
Backend->>Blockchain: Register Identity Hash
Blockchain->>Backend: Confirm Registration
Backend->>Frontend: Return Success
Frontend->>User: Display Dashboard

%% Content Access Flow

User->>Frontend: Request Course Material
Frontend->>Backend: Query Content Location
Backend->>P2P: Request Content Metadata
P2P->>Backend: Return Magnet URI/Torrent Info
Backend->>Frontend: Return Content Access Info
Frontend->>P2P: Connect to Torrent Network
P2P->>Frontend: Stream/Download Content
Frontend->>User: Display Content

%% Governance Flow

User->>Frontend: Create Governance Proposal
Frontend->>Backend: Submit Proposal
Backend->>Blockchain: Store Proposal
Blockchain->>Backend: Confirm Storage
Backend->>Frontend: Return Proposal ID

%% Voting Flow

User->>Frontend: Cast Vote
Frontend->>ZKSystem: Generate Anonymous Vote Proof
Frontend->>Backend: Submit Anonymous Vote
Backend->>ZKSystem: Verify Vote Validity
Backend->>Blockchain: Record Vote
Blockchain->>Backend: Confirm Vote Recording
Backend->>Frontend: Return Vote Confirmation
Frontend->>User: Display Vote Status

%% Reward Distribution Flow

Blockchain->>Blockchain: Execute Smart Contract
Blockchain->>Backend: Notify of Reward Event
Backend->>Frontend: Update User Balance
Frontend->>User: Display Reward Notification